

Learning objectives

In this lesson students will learn to:

- Draw and label a diagram of the inside of a computer, label each hardware component and briefly describe its role
- Explain how the speed of the clock impacts on performance
- Explain how pipelining improves the performance of the CPU.

For more details on this topic and additional student activities see Topic 3 of the student book.

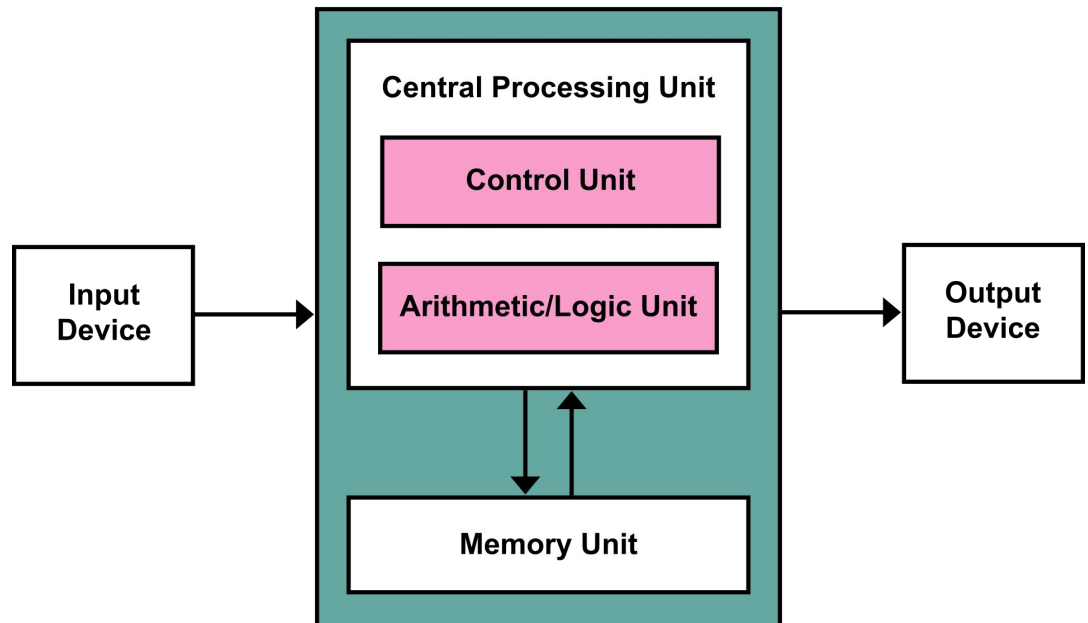
Recap

Instructions and data are stored in the memory (RAM).

Instructions are fetched one at a time into the processor.

The instructions are decoded by the control unit.

The instructions are executed, sometimes using the arithmetic logic unit

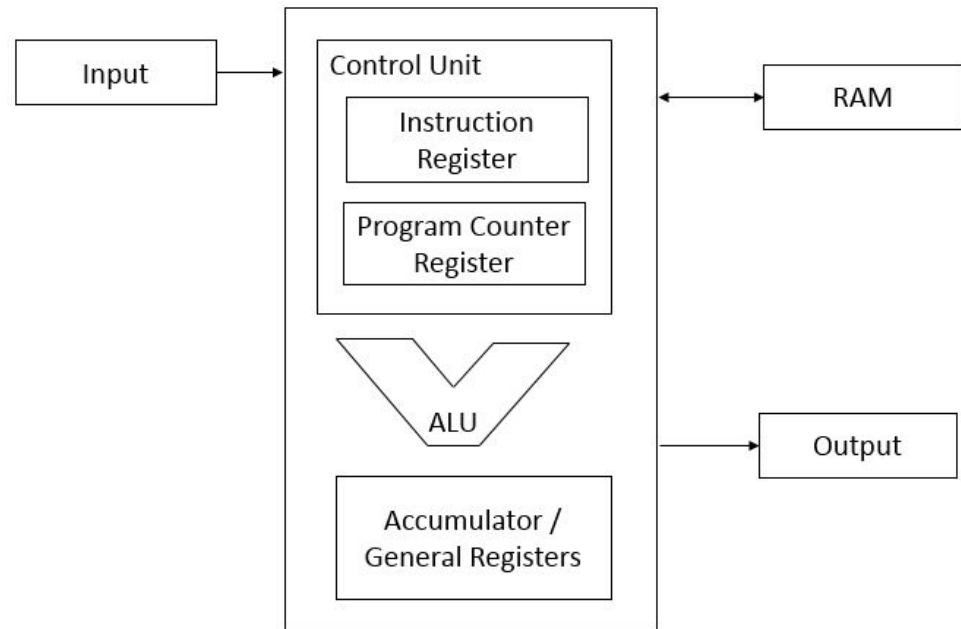


Central Processing Unit

The CPU is the 'brain of the computer'.

It is the hardware component responsible for all the processing that the computer carries out.

Its job is to process data. By processing, we mean things such as searching, sorting, calculating and decision making.



Whenever you are on working on your computer, it is the CPU that is at the centre of everything.

Control unit

The control unit receives signals from other parts of the computer system and sends signals to them.

It is responsible for handling hardware interrupts (e.g. inputs from a mouse, keyboard, network, etc.).

It is responsible for fetching, decoding, and executing instructions.

It is the powerhouse of the CPU – it has the most responsibility.

Arithmetic logic unit

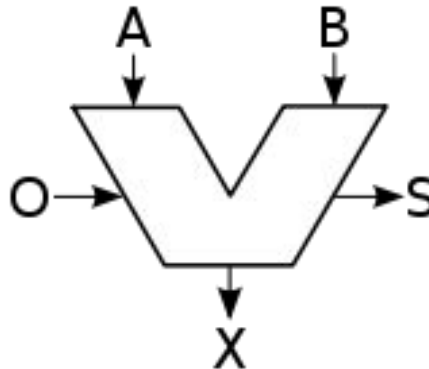
The Arithmetic Logic Unit (ALU) performs actual operations on data.

- Arithmetic: $+$ $/$ $=$ $*$
- Logic: AND, OR, NOT, etc.

It is able to compare numbers against 0.

It can test if two numbers are equal.

It uses logic gates in combination to perform operations.



Registers

Also known **as immediate access storage**.

Registers are a type of memory that is extremely fast, much faster than RAM.

Each type of processor has different registers that are designed to hold different information.

Most processors have:

- **an instruction register**: holds instruction currently being executed by the CPU.
- **an accumulator**: holds the accumulated result of operations carried out by the ALU.
- **a program counter**: holds the memory address of the next instruction to be executed.

Clock

A clock chip determines the speed of the CPU.

This chip uses a vibrating crystal that maintains a constant rate of vibration.

The speed of the clock is measured in Hertz (Hz), which is the amount of cycles per second.

A clock speed of 500Hz means 500 cycles per second.

Current computers have clock speeds of 3GHz, which means 3-billion cycles per second.

Each 'tick' means that one part of the fetch-decode-execute cycle can be carried out.

Pipelining

Each instruction that is carried out has several stages (fetch-decode-execute).

Each component of the CPU has a different job to do in this cycle.

Sometimes, one component has nothing to do whilst it is waiting for the component before it in the sequence to complete its task.

This increases the overall processing time needed.

Pipelining is a method of keeping all the components busy so as to improve overall CPU performance.

Pipelining

Pipelining attempts to keep every component of the CPU busy at all times.

Instructions flow through the CPU in stages.

This means that an instruction can be completed every clock cycle. This increases CPU **throughput** allowing overall performance to be optimised.

A CPU is called 'fully pipelined' if an instruction can be fetched every clock cycle.

Wrap up: you have learned how to...

- ✓ Draw and label a diagram of the inside of a computer, label each hardware component and briefly describe its role.
 - CPU contains the ALU, CU and registers.
 - The other components a computer needs include: input and output devices, RAM, motherboard, and power supply unit.
- ✓ Explain how the speed of the clock impacts on performance.
 - The faster the clock speed, the more instructions can be carried out per second.
- ✓ Explain how pipelining improves the performance of the CPU.
 - Pipelining reduces wasted clock cycles.
 - A fully pipelined CPU can fetch an instruction every clock tick.